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## References

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**Aalen et al.: Can we believe the DAGs? A comment on the relationship between causal DAGs and mechanisms** **Aalen-Roysland-Gran-etal-2016**

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Odd O. Aalen, Kjetil Røysland, Jon Michael Gran, Roger Kouyos, et al. “Can we believe the DAGs? A comment on the relationship between causal DAGs and mechanisms”. In: *Statistical Methods in Medical Research* 25.5 (July 2016), pp. 2294–2314. ISSN: 1477-0334. DOI: [10.1177/0962280213520436](https://doi.org/10.1177/0962280213520436).

Abstract: Directed acyclic graphs (DAGs) play a large role in the modern approach to causal inference. DAGs describe the relationship between measurements taken at various discrete times including the effect of interventions. The causal mechanisms, on the other hand, would naturally be assumed to be a continuous process operating over time in a cause-effect fashion. How does such immediate causation, that is causation occurring over very short time intervals, relate to DAGs constructed from discrete observations? We introduce a time-continuous model and simulate discrete observations in order to judge the relationship between the DAG and the immediate causal model. We find that there is no clear relationship; indeed the Bayesian network described by the DAG may not relate to the causal model. Typically, discrete observations of a process will obscure the conditional dependencies that are represented in the underlying mechanistic model of the process. It is therefore doubtful whether DAGs are always suited to describe causal relationships unless time is explicitly considered in the model. We relate the issues to mechanistic modeling by using the concept of local (in)dependence. An example using data from the Swiss HIV Cohort Study is presented.

Odd O. Aalen, Kjetil Røysland, Jon Michael Gran, and Bruno Ledergerber. “Causality, mediation and time: A dynamic viewpoint”. In: *Journal of the Royal Statistical Society. Series A (Statistics in Society)* 175.4 (2012), pp. 831–861. ISSN: 09641998, 1467985X. DOI: [10.1111/j.1467-985X.2011.01030.x](https://doi.org/10.1111/j.1467-985X.2011.01030.x).

Abstract: Time dynamics are often ignored in causal modelling. Clearly, causality must operate in time and we show how this corresponds to a mechanistic, or system, understanding of causality. The established counterfactual definitions of direct and indirect effects depend on an ability to manipulate the mediator which may not hold in practice, and we argue that a mechanistic view may be better. Graphical representations based on local independence graphs and dynamic path analysis are used to facilitate communication as well as providing an overview of the dynamic relations ‘at a glance’. The relationship between causality as understood in a mechanistic and in an interventionist sense is discussed. An example using data from the Swiss HIV Cohort Study is presented.

**Antonakis et al.: On ignoring the random effects assumption in multilevel models:  
Review, critique, and recommendations**

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**Antonakis-Bastardoz-Ronkko-2019**

John Antonakis, Nicolas Bastardoz, and Mikko Rönkkö. “On ignoring the random effects assumption in multilevel models: Review, critique, and recommendations”. In: *Organizational Research Methods* 24.2 (Oct. 2019), pp. 443–483. ISSN: 1552-7425. DOI: [10.1177/1094428119877457](https://doi.org/10.1177/1094428119877457).

Abstract: Entities such as individuals, teams, or organizations can vary systematically from one another. Researchers typically model such data using multilevel models, assuming that the random effects are uncorrelated with the regressors. Violating this testable assumption, which is often ignored, creates an endogeneity problem thus preventing causal interpretations. Focusing on

two-level models, we explain how researchers can avoid this problem by including cluster means of the Level 1 explanatory variables as controls; we explain this point conceptually and with a large-scale simulation. We further show why the common practice of centering the predictor variables is mostly unnecessary. Moreover, to examine the state of the science, we reviewed 204 randomly drawn articles from macro and micro organizational science and applied psychology journals, finding that only 106 articles—with a slightly higher proportion from macro-oriented fields—properly deal with the random effects assumption. Alarming, most models also failed on the usual exogeneity requirement of the regressors, leaving only 25 mostly macro-level articles that potentially reported trustworthy multilevel estimates. We offer a set of practical recommendations for researchers to model multilevel data appropriately.

**Appelbaum et al.: Journal article reporting standards for quantitative research in psychology: The APA Publications and Communications Board task force report**

**Appelbaum-Cooper-Kline-et-al-2018**

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Mark Appelbaum et al. “Journal article reporting standards for quantitative research in psychology: The APA Publications and Communications Board task force report”. In: *American Psychologist* 73.7 (Oct. 2018), pp. 947–947. ISSN: 0003-066X. DOI: [10.1037/amp0000389](https://doi.org/10.1037/amp0000389).

Abstract: Following a review of extant reporting standards for scientific publication, and reviewing 10 years of experience since publication of the first set of reporting standards by the American Psychological Association (APA; APA Publications and Communications Board Working Group on Journal Article Reporting Standards, 2008), the APA Working Group on Quantitative Research Reporting Standards recommended some modifications to the original standards. Examples of modifications include division of hypotheses, analyses, and conclusions into 3 groupings (primary, secondary, and exploratory) and some changes to the section on meta-analysis. Several new modules are included that report standards for observational studies, clinical trials, longitudinal studies, replication studies, and N-of-1 studies. In addition, standards for analytic methods with unique characteristics and output (structural equation modeling and Bayesian analysis) are in-

cluded. These proposals were accepted by the Publications and Communications Board of APA and supersede the standards included in the 6th edition of the Publication Manual of the American Psychological Association (APA, 2010).

**Asparouhov et al.: Dynamic structural equation models**

**Asparouhov-Hamaker-Muthen-2018**

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Tihomir Asparouhov, Ellen L. Hamaker, and Bengt Muthén. “Dynamic structural equation models”. In: *Structural Equation Modeling: A Multidisciplinary Journal* 25.3 (2018), pp. 359–388. DOI: [10.1080/10705511.2017.1406803](https://doi.org/10.1080/10705511.2017.1406803).

Abstract: This article presents dynamic structural equation modeling (DSEM), which can be used to study the evolution of observed and latent variables as well as the structural equation models over time. DSEM is suitable for analyzing intensive longitudinal data where observations from multiple individuals are collected at many points in time. The modeling framework encompasses previously published DSEM models and is a comprehensive attempt to combine time-series modeling with structural equation modeling. DSEM is estimated with Bayesian methods using the Markov chain Monte Carlo Gibbs sampler and the Metropolis-Hastings sampler. We provide a detailed description of the estimation algorithm as implemented in the Mplus software package. DSEM can be used for longitudinal analysis of any duration and with any number of observations across time. Simulation studies are used to illustrate the framework and study the performance of the estimation method. Methods for evaluating model fit are also discussed.

**Asparouhov et al.: Latent variable centering of predictors and mediators in multilevel and time-series models**

**Asparouhov-Muthen-2018**

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Tihomir Asparouhov and Bengt Muthén. “Latent variable centering of predictors and mediators in multilevel and time-series models”. In: *Structural Equation Modeling: A Multidisciplinary Journal* 26.1 (Sept. 2018), pp. 119–142. ISSN: 1532-8007. DOI: [10.1080/10705511.2018.1511375](https://doi.org/10.1080/10705511.2018.1511375).

**Bainter et al.: Advantages of integrative data analysis for developmental research**

**Bainter-Curran-2015**

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Sierra A. Bainter and Patrick J. Curran. “Advantages of integrative data analysis for developmental research”. In: *Journal of Cognition and Development* 16.1 (Jan. 2015), pp. 1–10. ISSN: 1532-7647. DOI: [10.1080/15248372.2013.871721](https://doi.org/10.1080/15248372.2013.871721).

Abstract: Amid recent progress in cognitive development research, high-quality data resources are accumulating, and data sharing and secondary data analysis are becoming increasingly valuable tools. Integrative data analysis (IDA) is an exciting analytical framework that can enhance secondary data analysis in powerful ways. IDA pools item-level data across multiple studies to make inferences possible both within and across studies and can be used to test questions not possible in individual contributing studies. Some of the potential benefits of IDA include the ability to study longer developmental periods, examine how the measurement of key constructs changes over time, increase subject heterogeneity, and improve statistical power and capability to study rare behaviors. Our goal in this article is to provide a brief overview of the benefits and challenges of IDA in developmental research and to identify additional resources that provide more detailed discussions of this topic.

**Barker et al.: Habitual alcohol seeking: Modeling the transition from casual drinking to addiction**

**Barker-Taylor-2014**

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Jacqueline M Barker and Jane R. Taylor. “Habitual alcohol seeking: Modeling the transition from casual drinking to addiction”. In: *Neuroscience & Biobehavioral Reviews* 47 (Nov. 2014), pp. 281–294. ISSN: 0149-7634. DOI: [10.1016/j.neubiorev.2014.08.012](https://doi.org/10.1016/j.neubiorev.2014.08.012).

Abstract: The transition from goal-directed actions to habitual ethanol seeking models the development of addictive behavior that characterizes alcohol use disorders. The progression to habitual ethanol-seeking behavior occurs more rapidly than for natural rewards, suggesting that

ethanol may act on habit circuit to drive the loss of behavioral flexibility. This review will highlight recent research that has focused on the formation and expression of habitual ethanol seeking, and the commonalities and distinctions between ethanol and natural reward-seeking habits, with the goal of highlighting important, understudied research areas that we believe will lead toward the development of novel treatment and prevention strategies for uncontrolled drinking.

**Barlow et al.: Toward a unified treatment for emotional disorders - republished article**

**Barlow-Allen-Choate-2016**

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David H. Barlow, Laura B. Allen, and Molly L. Choate. "Toward a unified treatment for emotional disorders - republished article". In: *Behavior Therapy* 47.6 (Nov. 2016), pp. 838–853. ISSN: 0005-7894. DOI: [10.1016/j.beth.2016.11.005](https://doi.org/10.1016/j.beth.2016.11.005).

Abstract: Over 40 years of development of cognitive behavioral approaches to treating anxiety and related emotional disorders have left us with highly efficacious treatments that are increasingly widely accepted. Nevertheless, these manualized protocols have become numerous and somewhat complex, restricting effective training and dissemination. Deepening understanding of the nature of emotional disorders reveals that commonalities in etiology and latent structure among these disorders supercedes differences. This suggests the possibility of distilling a set of psychological procedures that would comprise a unified intervention for emotional disorders. Based on theory and data emerging from the fields of learning, emotional development and regulation, and cognitive science, we identify three fundamental therapeutic components relevant to the treatment of emotional disorders generally. These three components include (a) altering antecedent cognitive reappraisals; (b) preventing emotional avoidance; and (c) facilitating action tendencies not associated with the emotion that is dysregulated. This treatment takes place in the context of provoking emotional expression (emotional exposure) through situational, internal, and somatic (interoceptive cues), as well as through standard mood-induction exercises, and differs from patient to patient only in the situational cues and exercises utilized. Theory and rationale supporting this new approach are described along with some preliminary experience with the protocol. This unified treatment may



yond political science to all multilevel datasets. However, omitted variables could still bias estimated higher-level variable effects; as with any model, care is required in interpretation.

**Beltz et al.: Bridging the nomothetic and idiographic approaches to the analysis of clinical data** **Beltz-Wright-Sprague-et-al-2016**

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Adriene M. Beltz et al. “Bridging the nomothetic and idiographic approaches to the analysis of clinical data”. In: *Assessment* 23.4 (May 2016), pp. 447–458. ISSN: 1552-3489. DOI: [10.1177/1073191116648209](https://doi.org/10.1177/1073191116648209).

Abstract: The nomothetic approach (i.e., the study of interindividual variation) dominates analyses of clinical data, even though its assumption of homogeneity across people and time is often violated. The idiographic approach (i.e., the study of intraindividual variation) is best suited for analyses of heterogeneous clinical data, but its person-specific methods and results have been criticized as unwieldy. Group iterative multiple model estimation (GIMME) combines the assets of the nomothetic and idiographic approaches by creating person-specific maps that contain a group-level structure. The maps show how intensively measured variables predict and are predicted by each other at different time scales. In this article, GIMME is introduced conceptually and mathematically, and then applied to an empirical data set containing the negative affect, detachment, disinhibition, and hostility composite ratings from the daily diaries of 25 individuals with personality pathology. Results are discussed with the aim of elucidating GIMME’s potential for clinical research and practice.

**Bernardo et al.: Pathways from collectivist coping to life satisfaction among Chinese: The roles of locus-of-hope** **Bernardo-Wang-Pesigan-et-al-2017**

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Allan B.I. Bernardo et al. “Pathways from collectivist coping to life satisfaction among Chinese: The roles of locus-of-hope”. In: *Personality and Individual Differences* 106 (Feb. 2017), pp. 253–256. ISSN: 0191-8869. DOI: [10.1016/j.paid.2016.10.059](https://doi.org/10.1016/j.paid.2016.10.059).



Abstract: Collectivist coping styles describe approaches to coping in collectivist cultures, but there is not much research on these coping styles and their relationship to well-being. We propose that two collectivistic coping styles (acceptance/reframing/striving and family support) contribute to life satisfaction by drawing from both personal and relational resources, and these resources are reflected in the distinct roles of internal and external-family loci-of-hope. Chinese students completed scales on collectivist coping, locus-of-hope, and life satisfaction. Path analysis of three models indicates that the most parsimonious model describes distinct pathways where acceptance/reframing/striving relates to life satisfaction through internal locus-of-hope and family support relates to life satisfaction through external-family locus-of-hope. The use of specific collectivist coping styles and reports of life satisfaction involves distinct personal and relational resources for coping that are valued in collectivist societies.

**Biesanz et al.: Assessing mediational models: Testing and interval estimation for indirect effects** **Biesanz-Falk-Savalei-2010**

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Jeremy C. Biesanz, Carl F. Falk, and Victoria Savalei. “Assessing mediational models: Testing and interval estimation for indirect effects”. In: *Multivariate Behavioral Research* 45.4 (Aug. 2010), pp. 661–701. DOI: [10.1080/00273171.2010.498292](https://doi.org/10.1080/00273171.2010.498292).

Abstract: Theoretical models specifying indirect or mediated effects are common in the social sciences. An indirect effect exists when an independent variable’s influence on the dependent variable is mediated through an intervening variable. Classic approaches to assessing such mediational hypotheses (Baron & Kenny, 1986; Sobel, 1982) have in recent years been supplemented by computationally intensive methods such as bootstrapping, the distribution of the product methods, and hierarchical Bayesian Markov chain Monte Carlo (MCMC) methods. These different approaches for assessing mediation are illustrated using data from Dunn, Biesanz, Human, and Finn (2007). However, little is known about how these methods perform relative to each other, particularly in more challenging situations, such as with data that are incomplete and/or nonnormal. This article presents an extensive Monte Carlo simulation evaluating a host of approaches for assessing media-

tion. We examine Type I error rates, power, and coverage. We study normal and nonnormal data as well as complete and incomplete data. In addition, we adapt a method, recently proposed in statistical literature, that does not rely on confidence intervals (CIs) to test the null hypothesis of no indirect effect. The results suggest that the new inferential method—the partial posterior p value—slightly outperforms existing ones in terms of maintaining Type I error rates while maximizing power, especially with incomplete data. Among confidence interval approaches, the bias-corrected accelerated (BCa) bootstrapping approach often has inflated Type I error rates and inconsistent coverage and is not recommended. In contrast, the bootstrapped percentile confidence interval and the hierarchical Bayesian MCMC method perform best overall, maintaining Type I error rates, exhibiting reasonable power, and producing stable and accurate coverage rates.

#### **Blanca et al.: Skewness and kurtosis in real data samples**

**Blanca-Arnau-LopezMontiel-et al-2013**

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Maria J. Blanca et al. “Skewness and kurtosis in real data samples”. In: *Methodology* 9.2 (May 2013), pp. 78–84. DOI: [10.1027/1614-2241/a000057](https://doi.org/10.1027/1614-2241/a000057).

Abstract: Parametric statistics are based on the assumption of normality. Recent findings suggest that Type I error and power can be adversely affected when data are non-normal. This paper aims to assess the distributional shape of real data by examining the values of the third and fourth central moments as a measurement of skewness and kurtosis in small samples. The analysis concerned 693 distributions with a sample size ranging from 10 to 30. Measures of cognitive ability and of other psychological variables were included. The results showed that skewness ranged between -2.49 and 2.33. The values of kurtosis ranged between -1.92 and 7.41. Considering skewness and kurtosis together the results indicated that only 5.5% of distributions were close to expected values under normality. Although extreme contamination does not seem to be very frequent, the findings are consistent with previous research suggesting that normality is not the rule with real data.

**Boden et al.: Are emotional clarity and emotion differentiation related?**

**Boden-Thompson-Dizen-et-al-2013**

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Matthew Tyler Boden et al. “Are emotional clarity and emotion differentiation related?” In: *Cognition and Emotion* 27.6 (Sept. 2013), pp. 961–978. ISSN: 1464-0600. DOI: [10.1080/02699931.2012.751899](https://doi.org/10.1080/02699931.2012.751899).

Abstract: Distinct literatures have developed regarding the constructs of emotional clarity (people’s meta-knowledge of their affective experience) and emotion differentiation (people’s ability to differentiate affective experience into discrete categories, e.g., anger vs. fear). Conceptually, emotion differentiation processes might be expected to contribute to increased emotional clarity. However, the relation between emotional clarity and emotion differentiation has not been directly investigated. In two studies with independent, undergraduate student samples, we measured emotional clarity using a self-report measure and derived emotion differentiation scores from scenario-based (Study 1) and event-sampling-based (Study 2) measures of affect. We found that emotional clarity and emotion differentiation are: (i) associated to a very small and statistically insignificant degree; and (ii) differentially associated with trait and scenario-based/event-sampling-based measures of affect intensity and variability. These results suggest that emotional clarity and differentiation are distinct constructs with unique relations to various facets of affective experience.

**Boettiger et al.: An introduction to Rocker: Docker containers for R**

**Boettiger-Eddelbuettel-2017**

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Carl Boettiger and Dirk Eddelbuettel. “An introduction to Rocker: Docker containers for R”. In: *The R Journal* 9.2 (2017), p. 527. DOI: [10.32614/rj-2017-065](https://doi.org/10.32614/rj-2017-065).

Abstract: We describe the Rocker project, which provides a widely-used suite of Docker images with customized R environments for particular tasks. We discuss how this suite is organized, and

how these tools can increase portability, scaling, reproducibility, and convenience of R users and developers.